MT-1004
Linear Algebra

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National University of Computer and Emerging Sciences

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Chapter 6

Orthogonality and Least Squares

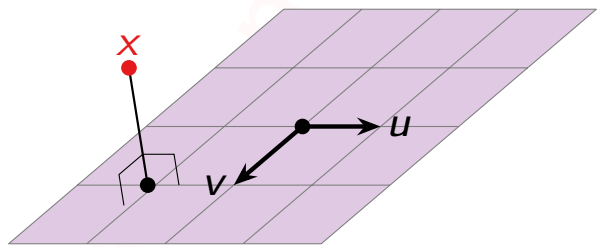
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Motivation

Recall: This course is about learning to:

- Solve the matrix equation $Ax = b$ (Echelon Form, Reduced Echelon Form, Column Space, Null Space)
- Solve the matrix equation $Ax = \lambda x$ (Eigenvalues, eigenvectors and their applications)
- Almost solve the equation $Ax = b$ (We are going to study)

Idea 1: In the real world, data is imperfect. Suppose you measure a data point x which you know for theoretical reasons must lie on a plane spanned by two vectors u and v.



Due to measurement error, though, the measured x is not actually in $\text{Span}\{u, v\}$. In other words, the equation $au + bv = x$ has no solution ($x \in$ Span of u and v). What do you do? The real value is probably the *closest* point to x on $\text{Span}\{u, v\}$. Which point is that?

The Dot Product

Purpose To find *length* and *angle* between two vectors, and in particular, to understand the notion of *orthogonality* (i.e. perpendicularity).

Definition

The **dot product** of two vectors x, y in $\mathbf{R}^n$ is

$$x \cdot y = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} := x_1y_1 + x_2y_2 + \cdots + x_ny_n = y^T x.$$

Example

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = 1 \cdot 4 + 2 \cdot 5 + 3 \cdot 6 = 32.$$

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Properties of the Dot Product

- $x \cdot x \geq 0$
- $x \cdot x = 0$ if and only if $x = 0$.
- $x \cdot y = y \cdot x$
- $(x + y) \cdot z = x \cdot z + y \cdot z$
- $(cx) \cdot y = c(x \cdot y)$

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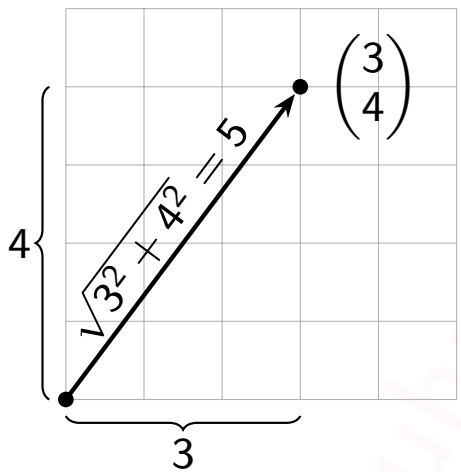
The Dot Product and Length

Definition

The **length** or **norm** of a vector x in $\mathbf{R}^n$ is

$$\|x\| = \sqrt{x \cdot x} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}.$$

The Pythagorean theorem!



$$\left\| \begin{pmatrix}3\\4\end{pmatrix} \right\| = \sqrt{3^2+4^2} = 5$$

Fact

If x is a vector and c is a scalar, then $\|cx\| = |c| \cdot \|x\|$.

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The Dot Product and Distance

Definition

The **distance** between two points x, y in $\mathbf{R}^n$ is

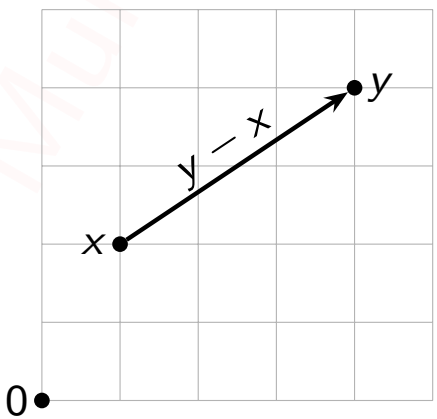
$$\text{dist}(x, y) = \|y - x\|.$$

This is just the length of the vector from x to y.

Example

Let $x = (1, 2)$ and $y = (4, 4)$. Then

$$\text{dist}(x, y) = \|y - x\| = \left\| \begin{pmatrix} 3 \\ 2 \end{pmatrix} \right\| = \sqrt{3^2 + 2^2} = \sqrt{13}.$$



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Unit Vectors

Definition

A **unit vector** is a vector v with length $\|v\| = 1$.

Example

The unit coordinate vectors are unit vectors:

$$\|e_1\| = \left\| \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\| = \sqrt{1^2 + 0^2 + 0^2} = 1$$

Definition

Let x be a nonzero vector in $\mathbf{R}^n$. The **unit vector in the direction of** x is the vector $\dfrac{x}{\|x\|}$.

This is in fact a unit vector:

scalar

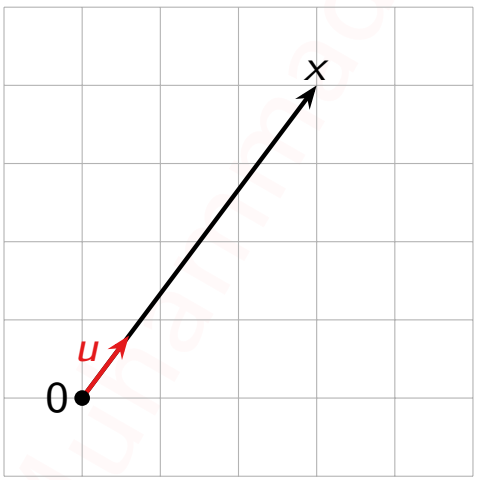
$$\left\| \frac{x}{\|x\|} \right\| = \frac{1}{\|x\|} \|x\| = 1.$$

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Example

What is the unit vector in the direction of $x = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$?

$$u = \frac{x}{\|x\|} = \frac{1}{\sqrt{3^2 + 4^2}} \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 3 \\ 4 \end{pmatrix}.$$



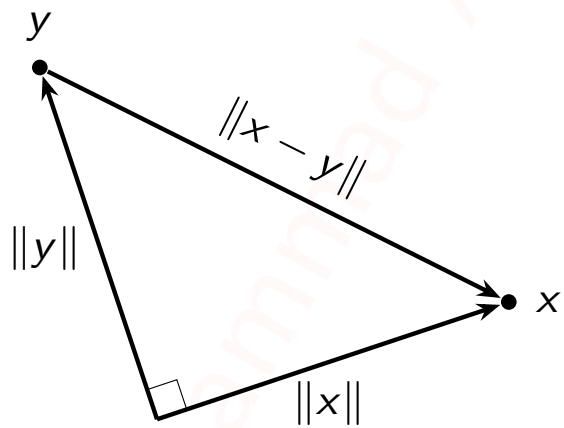
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Orthogonality

Definition

Two vectors x, y are **orthogonal** or **perpendicular** if $x \cdot y = 0$.
Notation: $x \perp y$ means $x \cdot y = 0$.

Why is this a good definition? The Pythagorean theorem / law of cosines!



$$
\begin{aligned}
x \text{ and } y \text{ are} \\
\text{perpendicular}
\end{aligned}
\quad \iff \|x\|^2 + \|y\|^2 = \|x - y\|^2
$$

$$\iff x \cdot x + y \cdot y = (x - y) \cdot (x - y)$$

$$\iff x \cdot x + y \cdot y = x \cdot x + y \cdot y - 2x \cdot y$$

$$\iff x \cdot y = 0$$

Fact: $x \perp y \iff \|x - y\|^2 = \|x\|^2 + \|y\|^2$

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Orthogonality

Definition

A set of vectors $\{x_1, x_2, ..., x_k\}$ in R^n is called an **orthogonal set** if $x_i \cdot x_j = 0$ whenever $i \neq j$ for $i, j = 1, 2, ..., k$.

Problem: Show that $\{x_1,\ x_2,\ x_3\}$ is an orthogonal set in $\mathbf{R}^3$ if

$$x_1 = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix},\ x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix},\ x_3 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

$$x_1 \cdot x_2 = 2(0) + 1(1) + (-1)(1) = 0$$
$$x_2 \cdot x_3 = 0(1) + 1(-1) + (1)(1) = 0$$
$$x_1 \cdot x_3 = 2(1) + 1(-1) + (-1)(1) = 0$$

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Orthogonality

Example

Problem: Find *all* vectors orthogonal to $v = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$.

We have to find all vectors x such that $x \cdot v = 0$. This means solving the equation

$$0 = x \cdot v = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = x_1 + x_2 - x_3.$$

The parametric form for the solution is $x_1 = -x_2 + x_3$, so the parametric vector form of the general solution is

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

For instance, $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \perp \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ because $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = 0.$

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Orthogonality

Example

Problem: Find *all* vectors orthogonal to both $v = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ and $w = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

Now we have to solve the system of two homogeneous equations

$$0 = x \cdot v = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = x_1 + x_2 - x_3$$

$$0 = x \cdot w = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = x_1 + x_2 + x_3.$$

In matrix form:

The rows are v and w $\longrightarrow \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix} \overset{\text{rref}}{\rightsquigarrow} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$.

The parametric vector form of the solution is

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}.$$

Orthogonal Basis

Theorem

If $\{u_1, u_2, ..., u_k\}$ is an orthogonal set of nonzero vectors in R^n, then these vectors are linearly independent.

Suppose $\{u_1, u_2, \ldots, u_m\}$ is orthogonal. We need to show that the equation

$$c_1 u_1 + c_2 u_2 + \cdots + c_m u_m = 0$$

has only the trivial solution $c_1 = c_2 = \cdots = c_m = 0$.

$$0 = u_1 \cdot (c_1 u_1 + c_2 u_2 + \cdots + c_m u_m) = c_1(u_1 \cdot u_1) + 0 + 0 + \cdots + 0.$$

Hence $c_1 = 0$. Similarly for the other c_i.

Definition

An **orthogonal basis** for a subspace W of $\mathbf{R}^n$ is a basis of W that is an orthogonal set.

Problem Show that $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ forms orthogonal basis of $\mathbf{R}^2$.

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Orthogonal Basis

Example

Problem Find an orthogonal basis for the subspace W of $\mathbf{R}^3$ given by

$$W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} : x - y + 2z = 0 \right\}.$$

We have already studied how to calculate the basis of W.

$$\begin{bmatrix} y - 2z \\ y \\ z \end{bmatrix} = y \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + z \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix},$$

so, $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$ forms basis of W

They are not orthogonal.

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Orthogonal Basis

We want to find a vector of W that is orthogonal to either $\begin{bmatrix}1\\1\\0\end{bmatrix}$ or $\begin{bmatrix}-2\\0\\1\end{bmatrix}$.

Let $w = \begin{bmatrix}x\\y\\z\end{bmatrix}$ be a vector of W and is orthogonal to $\begin{bmatrix}-2\\0\\1\end{bmatrix}$.

As $w \in W$, so we can write

$$x - y + 2z = 0.$$

Orthogonality condition implies

$$-2x + z = 0.$$

Solving above two equations, we get

$$y = 5x,\ z = 2x.$$

So,

$$w = x \begin{bmatrix}1\\5\\2\end{bmatrix}.$$

So, $\begin{bmatrix}-2\\0\\1\end{bmatrix}, \begin{bmatrix}1\\5\\2\end{bmatrix}$ forms orthogonal basis of W.

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Orthogonal Basis

Theorem

Let $\{x_1, x_2, ..., x_k\}$ be an orthogonal basis for a subspace W of $\mathbf{R}^n$ and let w be any vector in W. Then the unique scalars $c_1, c_2, ..., c_k$ such that

$$w = c_1x_1 + c_2x_2 + ... + c_kx_k$$

are given by

$$c_i = \frac{w \cdot x_i}{x_i \cdot x_i}, \quad i = 1, 2, ..., k.$$

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Orthogonal Basis

Example

Problem Show that $B = \left\{ v_1 = \begin{bmatrix} 4 \\ -2 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ forms basis of $\mathbf{R}^2$ and write coordinate vector of $w = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$ w.r.t B.

As $v_1 \cdot v_2 = (4).(1) + (2)(-2) = 0$, so, v_1 and v_2 are orthogonal and hence forms basis of $\mathbf{R}^2$.
Now, for the coordinate vector

$$w = c_1 v_1 + c_2 v_2.$$

Taking dot product with v_1, we get

$$w \cdot v_1 = c_1 v_1 \cdot v_1$$
$$c_1 = \frac{w \cdot v_1}{v_1 \cdot v_1}$$

Similarly, $c_2 = \dfrac{w \cdot v_2}{v_2 \cdot v_2}$.
As $v_1 \cdot v_1 = 12,\ v_2 \cdot v_2 = 5,\ w \cdot v_1 = 10,\ w \cdot v_2 = -5$. So, $c_1 = \frac{5}{6},\ c_2 = -1$.
Hence, coordinate vector of w is

$$\begin{bmatrix} \frac{5}{6} \\ -1 \end{bmatrix}.$$

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Orthogonality

General procedure

Problem: Find all vectors orthogonal to some number of vectors $v_1, v_2, \ldots, v_m$ in $\mathbf{R}^n$.

This is the same as finding all vectors x such that

$$0 = v_1^T x = v_2^T x = \cdots = v_m^T x.$$

Putting the *row* vectors $v_1^T, v_2^T, \ldots, v_m^T$ into a matrix, this is the same as finding all x such that

$$\begin{pmatrix} — & v_1^T & — \\ — & v_2^T & — \\ & \vdots & \\ — & v_m^T & — \end{pmatrix} x = \begin{pmatrix} v_1 \cdot x \\ v_2 \cdot x \\ \vdots \\ v_m \cdot x \end{pmatrix} = 0.$$

Important

The set of all vectors orthogonal to some vectors $v_1, v_2, \ldots, v_m$ in $\mathbf{R}^n$ is the *null space* of the $m \times n$ matrix

$$\begin{pmatrix} — & v_1^T & — \\ — & v_2^T & — \\ & \vdots & \\ — & v_m^T & — \end{pmatrix}.$$

In particular, this set is a subspace!

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Orthogonal Complements

Definition

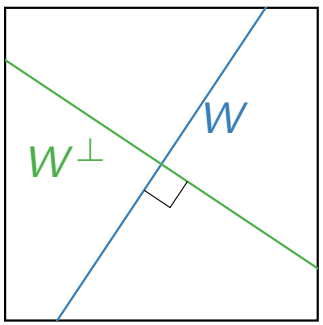
Let W be a subspace of $\mathbf{R}^n$. Its **orthogonal complement** is

$$W^\perp = \{v \text{ in } \mathbf{R}^n \mid v \cdot w = 0 \text{ for all } w \text{ in } W\} \qquad \text{read "}W \text{ perp"}.$$

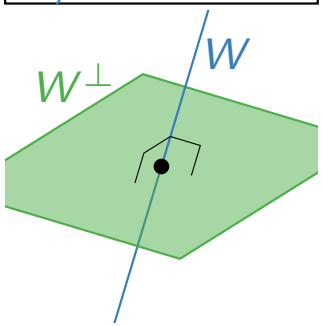
$W^\perp$ is orthogonal complement
A^T is transpose

Pictures:

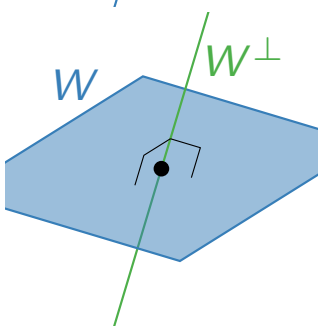
The orthogonal complement of a line in $\mathbf{R}^2$ is the perpendicular line.



The orthogonal complement of a line in $\mathbf{R}^3$ is the perpendicular plane.



The orthogonal complement of a plane in $\mathbf{R}^3$ is the perpendicular line.



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Poll

Poll

Let W be a plane in $\mathbf{R}^4$. How would you describe $W^\perp$?

A. The zero space $\{0\}$.

B. A line in $\mathbf{R}^4$.

C. A plane in $\mathbf{R}^4$.

D. A 3-dimensional space in $\mathbf{R}^4$.

E. All of $\mathbf{R}^4$.

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Orthogonal Complements

Basic properties

Let W be a subspace of $\mathbf{R}^n$.

Facts:

1. $W^\perp$ is also a subspace of $\mathbf{R}^n$
2. $(W^\perp)^\perp = W$
3. $\dim W + \dim W^\perp = n$
4. If $W = \text{Span}\{v_1, v_2, \ldots, v_m\}$, then

$$\begin{aligned} W^\perp &= \text{all vectors orthogonal to each } v_1, v_2, \ldots, v_m \\ &= \{x \text{ in } \mathbf{R}^n \mid x \cdot v_i = 0 \text{ for all } i = 1, 2, \ldots, m\} \\ &= \text{Nul} \begin{pmatrix} — & v_1^T & — \\ — & v_2^T & — \\ & \vdots & \\ — & v_m^T & — \end{pmatrix}. \end{aligned}$$

Let's check 1.

- Is 0 in $W^\perp$? Yes: $0 \cdot w = 0$ for any w in W.
- Suppose x, y are in $W^\perp$. So $x \cdot w = 0$ and $y \cdot w = 0$ for all w in W. Then $(x + y) \cdot w = x \cdot w + y \cdot w = 0 + 0 = 0$ for all w in W. So $x + y$ is also in $W^\perp$.
- Suppose x is in $W^\perp$. So $x \cdot w = 0$ for all w in W. If c is a scalar, then $(cx) \cdot w = c(x \cdot 0) = c(0) = 0$ for any w in W. So cx is in $W^\perp$.

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Orthogonal Complements

Computation

Problem: if $W = \text{Span}\left\{ \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$, compute $W^{\perp}$.

By property 4, we have to find the null space of the matrix whose rows are $\begin{pmatrix} 1 & 1 & -1 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}$, which we did before:

$$\text{Nul}\begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix} = \text{Span}\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

$$\text{Span}\{v_1, v_2, \ldots, v_m\}^{\perp} = \text{Nul}\begin{pmatrix} — & v_1^T & — \\ — & v_2^T & — \\ & \vdots & \\ — & v_m^T & — \end{pmatrix}$$

Orthogonal Complements

Row space, column space, null space

Definition

The **row space** of an $m \times n$ matrix A is the span of the *rows* of A. It is denoted $\operatorname{Row} A$. Equivalently, it is the column span of A^T:

$$\operatorname{Row} A = \operatorname{Col} A^T.$$

It is a subspace of $\mathbf{R}^n$.

We showed before that if A has rows $v_1^T, v_2^T, \ldots, v_m^T$, then

$$\operatorname{Span}\{v_1, v_2, \ldots, v_m\}^\perp = \operatorname{Nul} A.$$

Hence we have shown:

Fact: $(\operatorname{Row} A)^\perp = \operatorname{Nul} A$.

Replacing A by A^T, and remembering $\operatorname{Row} A^T = \operatorname{Col} A$:

Fact: $(\operatorname{Col} A)^\perp = \operatorname{Nul} A^T$.

Using property 2 and taking the orthogonal complements of both sides, we get:

Fact: $(\operatorname{Nul} A)^\perp = \operatorname{Row} A$ and $\operatorname{Col} A = (\operatorname{Nul} A^T)^\perp$.

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Orthogonal Complements

Reference sheet

Orthogonal Complements of Most of the Subspaces We've Seen

For any vectors $v_1, v_2, \ldots, v_m$:

$$\text{Span}\{v_1, v_2, \ldots, v_m\}^\perp = \text{Nul} \begin{pmatrix} — & v_1^T & — \\ — & v_2^T & — \\ & \vdots & \\ — & v_m^T & — \end{pmatrix}$$

For any matrix A:

$$\text{Row } A = \text{Col } A^T$$

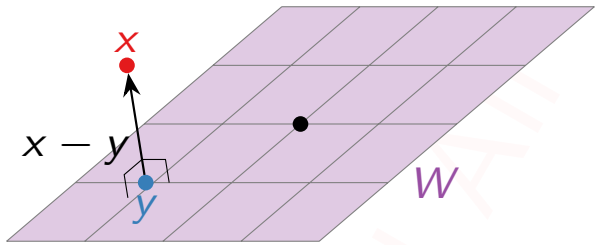
and

$$(\text{Row } A)^\perp = \text{Nul } A \qquad \text{Row } A = (\text{Nul } A)^\perp$$
$$(\text{Col } A)^\perp = \text{Nul } A^T \qquad \text{Col } A = (\text{Nul } A^T)^\perp$$

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Best Approximation

Suppose you measure a data point x which you know for theoretical reasons must lie on a subspace W.



Due to measurement error, though, the measured x is not actually in W. Best approximation: y is the *closest* point to x on W.

How do you know that y is the closest point? The vector from y to x is orthogonal to W: it is in the *orthogonal complement* $W^{\perp}$.

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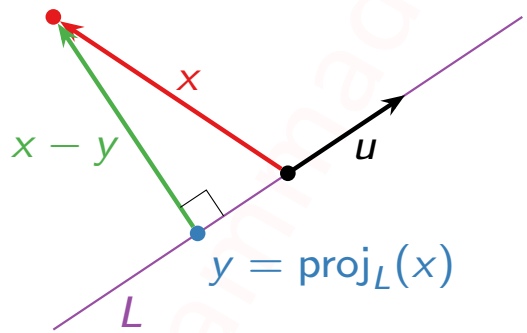
Orthogonal Projection onto a Line

Theorem

Let $L = \text{Span}\{u\}$ be a line in $\mathbf{R}^n$, and let x be in $\mathbf{R}^n$. The closest point to x on L is the point

$$\text{proj}_L(x) = \frac{x \cdot u}{u \cdot u}\, u.$$

This point is called the **orthogonal projection of x onto L.**



Why? Let $y = \text{proj}_L(x)$. We have to verify that $x - y$ is in $L^\perp$. This means proving that $u \cdot (x - y) = 0$.

$$u \cdot (x - y) = u \cdot \left(x - \frac{x \cdot u}{u \cdot u} u\right) = u \cdot x - \frac{x \cdot u}{u \cdot u}(u \cdot u) = u \cdot x - x \cdot u = 0.$$

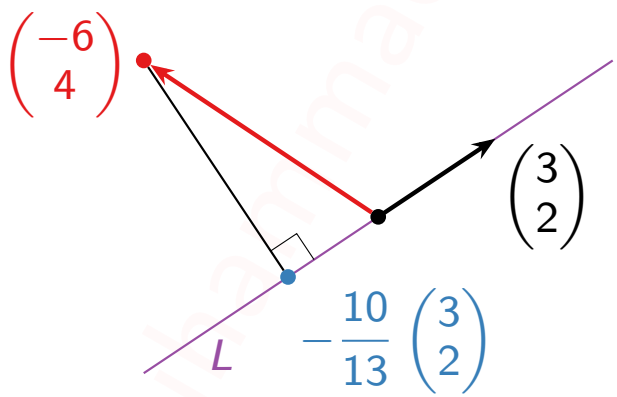
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Orthogonal Projection onto a Line

Example

Compute the orthogonal projection of $x = \begin{pmatrix} -6 \\ 4 \end{pmatrix}$ onto the line L spanned by $u = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$.

$$y = \text{proj}_L(x) = \frac{x \cdot u}{u \cdot u} u = \frac{-18 + 8}{9 + 4} \begin{pmatrix} 3 \\ 2 \end{pmatrix} = -\frac{10}{13} \begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$



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Orthogonal Sets

Definition

A set of *nonzero* vectors is **orthogonal** if each pair of vectors is orthogonal. It is **orthonormal** if, in addition, each vector is a unit vector.

Example: $\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$ is an orthogonal set. Check:

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} = 0 \qquad \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = 0 \qquad \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = 0.$$

Lemma

An orthogonal set of vectors is linearly independent.

Suppose $\{u_1, u_2, \ldots, u_m\}$ is orthogonal. We need to show that the equation

$$c_1 u_1 + c_2 u_2 + \cdots + c_m u_m = 0$$

has only the trivial solution $c_1 = c_2 = \cdots = c_m = 0$.

$$0 = u_1 \cdot (c_1 u_1 + c_2 u_2 + \cdots + c_m u_m) = c_1 (u_1 \cdot u_1) + 0 + 0 + \cdots + 0.$$

Hence $c_1 = 0$. Similarly for the other c_i.

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Orthogonal Bases

An orthogonal set $\mathcal{B} = \{u_1, u_2, \ldots, u_m\}$ forms a basis for $W = \operatorname{Span} \mathcal{B}$.

An advantage of orthogonal bases is it's *very easy* to compute the $\mathcal{B}$-coordinates of a vector in W.

Theorem

Let $\mathcal{B} = \{u_1, u_2, \ldots, u_m\}$ be an orthogonal set, and let x be a vector in $W = \operatorname{Span} \mathcal{B}$. Then

$$x = \sum_{i=1}^{m} \frac{x \cdot u_i}{u_i \cdot u_i} u_i = \frac{x \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{x \cdot u_2}{u_2 \cdot u_2} u_2 + \cdots + \frac{x \cdot u_m}{u_m \cdot u_m} u_m.$$

In other words, the $\mathcal{B}$-coordinates of x are $\left(\frac{x \cdot u_1}{u_1 \cdot u_1}, \frac{x \cdot u_2}{u_2 \cdot u_2}, \ldots, \frac{x \cdot u_m}{u_m \cdot u_m} \right)$.

Why? If $x = c_1 u_1 + c_2 u_2 + \cdots + c_m u_m$, then

$$x \cdot u_1 = c_1 (u_1 \cdot u_1) + 0 + \cdots + 0 \implies c_1 = \frac{x \cdot u_1}{u_1 \cdot u_1}.$$

Similarly for the other c_i.

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Orthogonal Bases

Geometric reason

Theorem

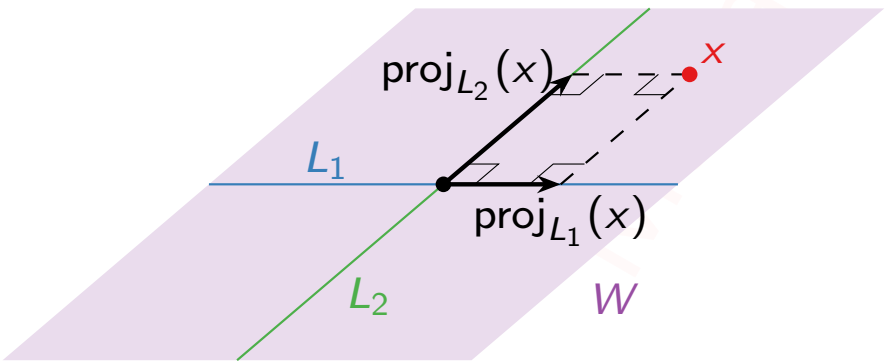
Let $\mathcal{B} = \{u_1, u_2, \ldots, u_m\}$ be an orthogonal set, and let x be a vector in $W = \operatorname{Span} \mathcal{B}$. Then

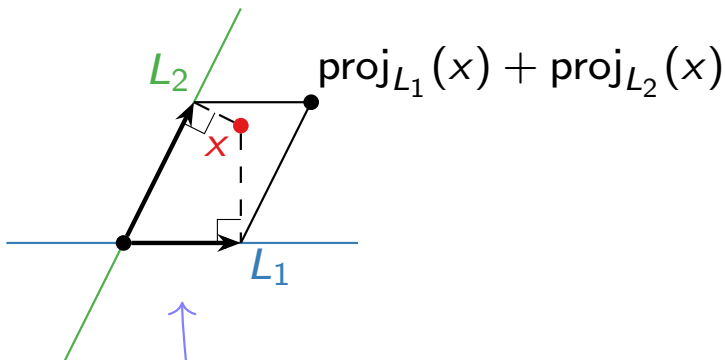
$$x = \sum_{i=1}^{m} \frac{x \cdot u_i}{u_i \cdot u_i} u_i = \frac{x \cdot u_1}{u_1 \cdot u_1} u_1 + \boxed{\frac{x \cdot u_2}{u_2 \cdot u_2} u_2} + \cdots + \frac{x \cdot u_m}{u_m \cdot u_m} u_m.$$

$\operatorname{proj}_{L_2}(u_2)$

If L_i is the line spanned by u_i, then this says

$$x = \operatorname{proj}_{L_1}(x) + \operatorname{proj}_{L_2}(x) + \cdots + \operatorname{proj}_{L_m}(x).$$





Warning: This only works for an *orthogonal* basis.

Orthogonal Bases

Example

Problem: Find the $\mathcal{B}$-coordinates of $x = \begin{pmatrix} 0 \\ 3 \end{pmatrix}$, where

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -4 \\ 2 \end{pmatrix} \right\}.$$

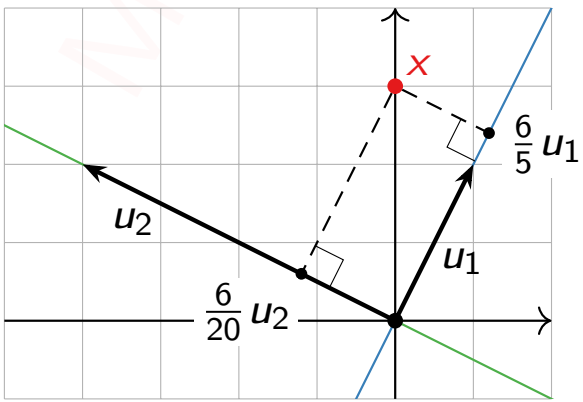
Old way:

$$\left(\begin{array}{cc|c} 1 & -4 & 0 \\ 2 & 2 & 3 \end{array}\right) \overset{\text{rref}}{\rightsquigarrow} \left(\begin{array}{cc|c} 1 & 0 & 6/5 \\ 0 & 1 & 6/20 \end{array}\right) \implies [x]_{\mathcal{B}} = \begin{pmatrix} 6/5 \\ 6/20 \end{pmatrix}.$$

New way: note $\mathcal{B}$ is an *orthogonal* basis.

$$x = \frac{x \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{x \cdot u_2}{u_2 \cdot u_2} u_2 = \frac{3 \cdot 2}{1^2 + 2^2} u_1 + \frac{3 \cdot 2}{(-4)^2 + 2^2} u_2 = \frac{6}{5} u_1 + \frac{6}{20} u_2.$$

So the $\mathcal{B}$-coordinates are $\frac{6}{5}, \frac{6}{20}$.



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Orthogonal Bases

Example

Problem: Find the $\mathcal{B}$-coordinates of $x = (6, 1, -8)$ where

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}.$$

Answer:

$$\begin{aligned}
[x]_{\mathcal{B}} &= \left(\frac{x \cdot u_1}{u_1 \cdot u_1}, \frac{x \cdot u_2}{u_2 \cdot u_2}, \frac{x \cdot u_3}{u_3 \cdot u_3} \right) \\
&= \left(\frac{6 \cdot 1 + 1 \cdot 1 - 8 \cdot 1}{1^2 + 1^2 + 1^2}, \frac{6 \cdot 1 + 1 \cdot (-2) - 8 \cdot 1}{1^2 + (-2)^2 + 1^2}, \frac{6 \cdot 1 + 1 \cdot 0 + (-8) \cdot (-1)}{1^2 + 0^2 + (-1)^2} \right) \\
&= \left(-\frac{1}{3}, -\frac{2}{3}, 7 \right).
\end{aligned}$$

Check:

$$\begin{pmatrix} 6 \\ 1 \\ -8 \end{pmatrix} = -\frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{2}{3} \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + 7 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}. \quad ✔$$

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Orthonormal Basis

Definition

A set of vectors in $\mathbf{R}^n$ is called an **orthonormal set** if it is an orthogonal set of unit vectors.

Theorem

Let $\{x_1, x_2, ..., x_k\}$ be an orthonormal basis for a subspace W of $\mathbf{R}^n$ and let w be any vector in W. Then the unique scalars $c_1, c_2, ..., c_k$ such that

$$w = c_1x_1 + c_2x_2 + ... + c_kx_k$$

are given by

$$c_i = w \cdot x_i, \quad i = 1, 2, ..., k.$$

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Orthogonal Matrices

Theorem

The matrix Q (square or rectangular) has orthonormal columns if and only if $Q^T Q = I$

Proof.

If Q has orthonormal columns then,

$$(Q^T Q)_{ij} = q_i \cdot q_j = I.$$

Conversely,
If $Q^T Q = I$, then

$$q_i \cdot q_j = \begin{cases} 1, & \text{if } i \neq j \\ 0, & \text{if} i = j. \end{cases}$$

□

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Orthogonal Matrices

Theorem

Any square matrix Q whose columns form an orthonormal set is called **Orthogonal Matrix**.

Theorem

Let Q be an $n \times n$ matrix. Then the following statements are equivalent:

1. Q is orthogonal.
2. $Q^T = Q^{-1}$.
3. $\|Q\mathbf{x}\| = \|\mathbf{x}\|$.
4. $Q\mathbf{x} \cdot Q\mathbf{y} = \mathbf{x} \cdot \mathbf{y}$.

   where $\mathbf{x}$ and $\mathbf{y}$ are from $\mathbf{R}^n$.

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Orthogonal Matrices

Theorem

Let Q be an orthogonal matrix.

1. Q^{-1} is orthogonal.
2. $det(Q) = \pm 1$
3. If λ is an eigenvalue of Q, then $|\lambda| = 1$.
4. Product of orthogonal matrices of same size is another orthogonal matrix.
5. Rows of Q forms an orthonormal set.

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Orthogonal Matrices

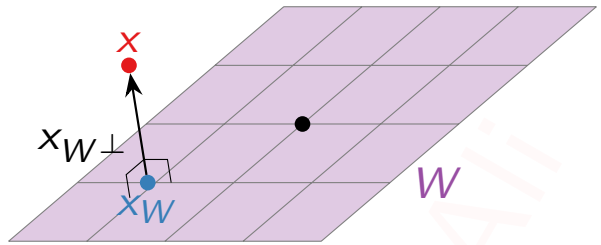
Examples

- $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & cos\theta \end{bmatrix}$.
- $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

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Idea Behind Orthogonal Projections

If x is not in a subspace W, then y in W is the closest to x if $x - y$ is in $W^\perp$:



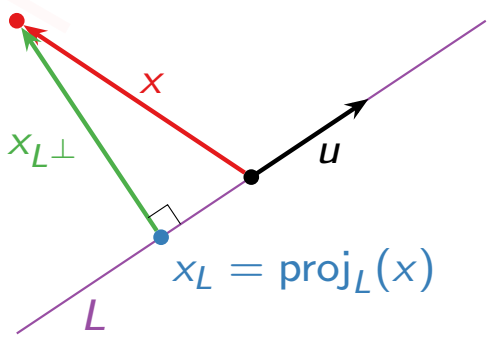
Reformulation: Every vector x can be decompsed uniquely as

$$x = x_W + x_{W^\perp}$$

where $x_W = y$ is the closest vector to x in W, and $x_{W^\perp} = x - y$ is in $W^\perp$.

Example: Let $u = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and let $L = \text{Span}\{u\}$. Let $x = \begin{pmatrix} -6 \\ 4 \end{pmatrix}$. Then the closest point to x in L is $\text{proj}_L(x) = \frac{x \cdot u}{u \cdot u} u$, so

$$x_L = \text{proj}_L(x) = -\frac{10}{13}\begin{pmatrix} 3 \\ 2 \end{pmatrix} \qquad x_{L^\perp} = x - \text{proj}_L(x) = \begin{pmatrix} -6 \\ 4 \end{pmatrix} + \frac{10}{13}\begin{pmatrix} 3 \\ 2 \end{pmatrix}.$$



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Orthogonal Projections

Definition

Let W be a subspace of $\mathbf{R}^n$, and let $\{u_1, u_2, \ldots, u_m\}$ be an *orthogonal* basis for W. The **orthogonal projection** of a vector x onto W is

$$\text{proj}_W(x) \stackrel{\text{def}}{=} \sum_{i=1}^{m} \frac{x \cdot u_i}{u_i \cdot u_i} u_i.$$

Question: What is the difference between this and the formula for $[x]_{\mathcal{B}}$ from before?

Theorem

Let W be a subspace of $\mathbf{R}^n$, and let x be a vector in $\mathbf{R}^n$. Then $\text{proj}_W(x)$ is the closest point to x in W. Therefore

$$x_W = \text{proj}_W(x) \qquad x_{W^\perp} = x - \text{proj}_W(x).$$

Why? Let $y = \text{proj}_W(x)$. We need to show that $x - y$ is in $W^\perp$. In other words, $u_i \cdot (x - y) = 0$ for each i. Let's do u_1:

$$u_1 \cdot (x - y) = u_1 \cdot \left(x - \sum_{i=1}^{m} \frac{x \cdot u_i}{u_i \cdot u_i} u_i \right) = u_1 \cdot x - \frac{x \cdot u_1}{u_1 \cdot u_1}(u_1 \cdot u_1) - 0 - \cdots = 0.$$

Orthogonal Projections

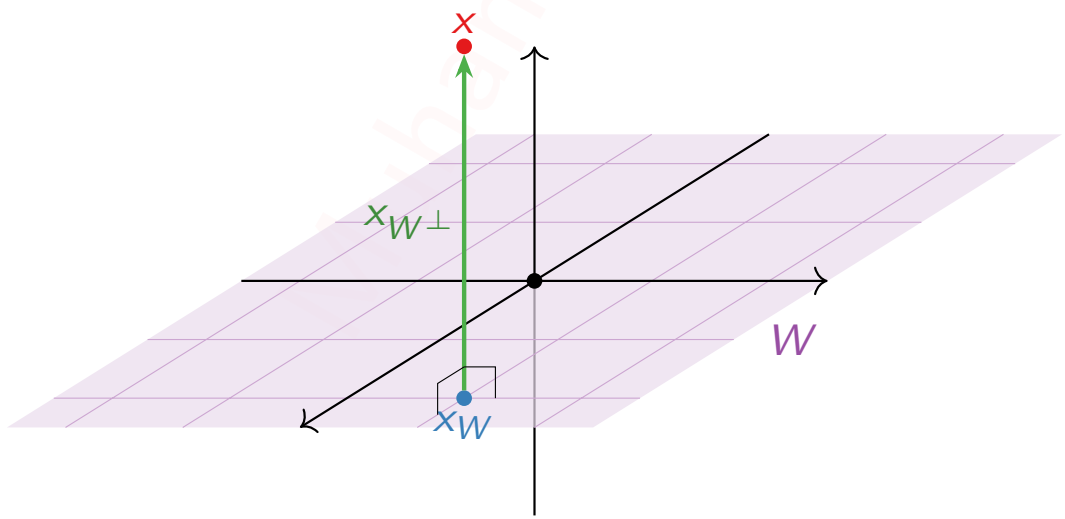
Easy example

What is the projection of $x = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ onto the xy-plane?

Answer: The xy-plane is $W = \text{Span}\{e_1, e_2\}$, and $\{e_1, e_2\}$ is an orthogonal basis.

$$x_W = \text{proj}_W \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \frac{x \cdot e_1}{e_1 \cdot e_1} e_1 + \frac{x \cdot e_2}{e_2 \cdot e_2} e_2 = \frac{1 \cdot 1}{1^2} e_1 + \frac{1 \cdot 2}{1^2} e_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}.$$

So this is the same projection as before.



Orthogonal Projections

More complicated example

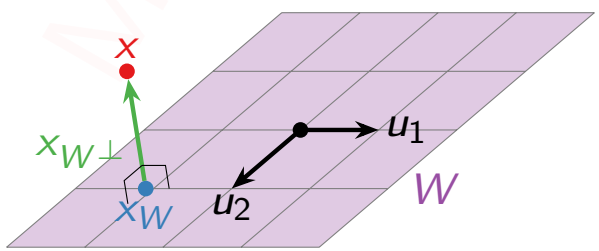
What is the projection of $x = \begin{pmatrix} -1.1 \\ 1.4 \\ 1.45 \end{pmatrix}$ onto $W = \text{Span}\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1.1 \\ -.2 \end{pmatrix} \right\}$?

Answer: The basis is orthogonal, so

$$x_W = \text{proj}_W \begin{pmatrix} -1.1 \\ 1.4 \\ 1.45 \end{pmatrix} = \frac{x \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{x \cdot u_2}{u_2 \cdot u_2} u_2$$

$$= \frac{(-1.1)(1)}{1^2} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \frac{(1.4)(1.1) + (1.45)(-.2)}{1.1^2 + (-.2)^2} \begin{pmatrix} 0 \\ 1.1 \\ -.2 \end{pmatrix}$$

This turns out to be equal to $u_2 - 1.1u_1$.



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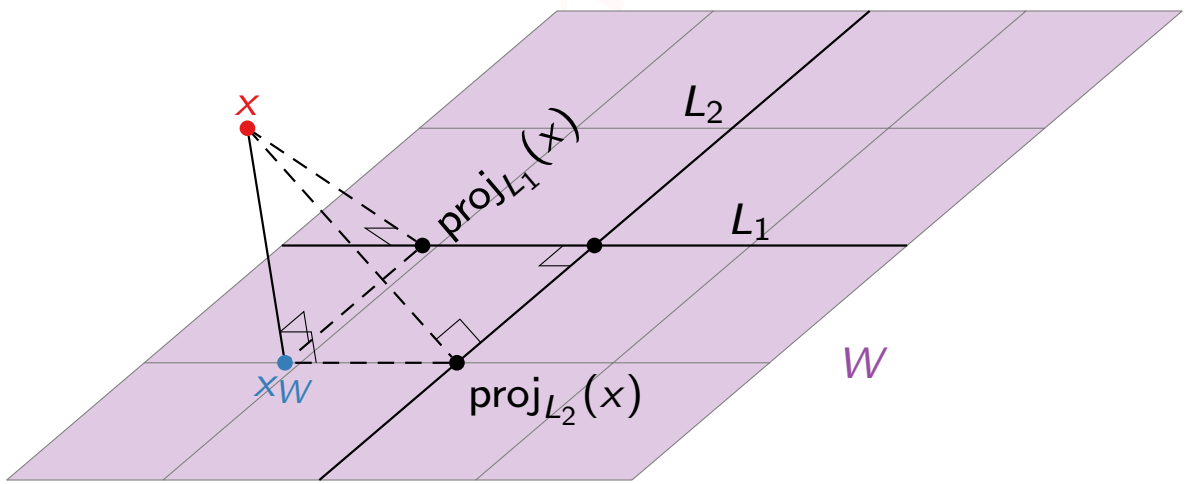
Orthogonal Projections

Picture

Let W be a subspace of $\mathbf{R}^n$, and let $\{u_1, u_2, \ldots, u_m\}$ be an orthogonal basis for W. Let $L_i = \text{Span}\{u_i\}$. Then

$$\text{proj}_W(x) = \sum_{i=1}^{m} \frac{x \cdot u_i}{u_i \cdot u_i} u_i = \sum_{i=1}^{m} \text{proj}_{L_i}(x).$$

So the orthogonal projection is formed by adding orthogonal projections onto perpendicular lines.



Orthogonal Projections

Properties

First we restate the property we've been using all along.

Best Approximation Theorem

Let W be a subspace of $\mathbf{R}^n$, and let x be a vector in $\mathbf{R}^n$. Then $y = \text{proj}_W(x)$ is the closest point in W to x, in the sense that

$$\text{dist}(x, y') \geq \text{dist}(x, y) \quad \text{for all} \quad y' \text{ in } W.$$

We can think of orthogonal projection as a *transformation*:

$$\text{proj}_W : \mathbf{R}^n \longrightarrow \mathbf{R}^n \qquad x \mapsto \text{proj}_W(x).$$

Theorem

Let W be a subspace of $\mathbf{R}^n$.

1. proj_W is a *linear* transformation.
2. For every x in W, we have $\text{proj}_W(x) = x$.
3. For every x in $W^\perp$, we have $\text{proj}_W(x) = 0$.
4. The range of proj_W is W.

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Poll

Let W be a subspace of $\mathbf{R}^n$.

Poll

Let A be the matrix for proj_W. What is/are the eigenvalue(s) of A?

A.0 B.1 C. -1 D.0, 1 E.1, -1 F.0, -1 G. -1, 0, 1

The 1-eigenspace is W.

The 0-eigenspace is $W^\perp$.

We have $\dim W + \dim W^\perp = n$, so that gives n linearly independent eigenvectors already.

So the answer is D.

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Orthogonal Projections

Matrices

What is the matrix for $\text{proj}_W : \mathbf{R}^3 \to \mathbf{R}^3$, where

$$W = \text{Span}\left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}?$$

Answer: Recall how to compute the matrix for a linear transformation:

$$A = \begin{pmatrix} | & | & | \\ \text{proj}_W(e_1) & \text{proj}_W(e_2) & \text{proj}_W(e_3) \\ | & | & | \end{pmatrix}.$$

We compute:

$$\text{proj}_W(e_1) = \frac{e_1 \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{e_1 \cdot u_2}{u_2 \cdot u_2} u_2 = \frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 5/6 \\ 1/3 \\ -1/6 \end{pmatrix}$$

$$\text{proj}_W(e_2) = \frac{e_2 \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{e_2 \cdot u_2}{u_2 \cdot u_2} u_2 = 0 + \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \end{pmatrix}$$

$$\text{proj}_W(e_3) = \frac{e_3 \cdot u_1}{u_1 \cdot u_1} u_1 + \frac{e_3 \cdot u_2}{u_2 \cdot u_2} u_2 = -\frac{1}{2} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/6 \\ 1/3 \\ 5/6 \end{pmatrix}$$

Therefore $A = \begin{pmatrix} 5/6 & 1/3 & -1/6 \\ 1/3 & 1/3 & 1/3 \\ -1/6 & 1/3 & 5/6 \end{pmatrix}$.

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Orthogonal Projections

Matrix facts

Let W be an m-dimensional subspace of $\mathbf{R}^n$, let $\text{proj}_W \colon \mathbf{R}^n \to W$ be the projection, and let A be the matrix for proj_L.

Fact 1: A is diagonalizable with eigenvalues 0 and 1; it is similar to the diagonal matrix with m ones and $n - m$ zeros on the diagonal.

Why? Let $v_1, v_2, \ldots, v_m$ be a basis for W, and let $v_{m+1}, v_{m+2}, \ldots, v_n$ be a basis for $W^\perp$. These are (linearly independent) eigenvectors with eigenvalues 1 and 0, respectively, and they form a basis for $\mathbf{R}^n$ because there are n of them.

Example: If W is a plane in $\mathbf{R}^3$, then A is similar to projection onto the xy-plane:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Fact 2: $A^2 = A$.

Why? Projecting twice is the same as projecting once:

$$\text{proj}_W \circ \text{proj}_W = \text{proj}_W \implies A \cdot A = A.$$

Orthogonal Projections

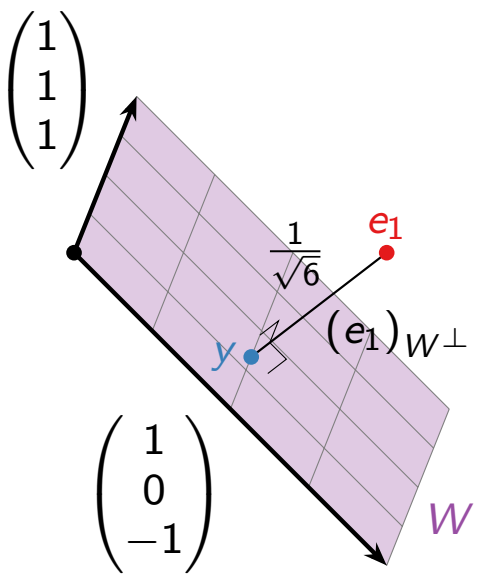
Minimum distance

What is the distance from e_1 to $W = \text{Span}\left\{\begin{pmatrix}1\\0\\-1\end{pmatrix}, \begin{pmatrix}1\\1\\1\end{pmatrix}\right\}$?

Answer: The closest point on W to e_1 is $\text{proj}_W(e_1) = \begin{pmatrix}5/6\\1/3\\-1/6\end{pmatrix}$.

The distance from e_1 to this point is

$$\begin{aligned}
\text{dist}\big(e_1, \text{proj}_W(e_1)\big) &= \|(e_1)_{W^\perp}\| \\
&= \left\| \begin{pmatrix}1\\0\\0\end{pmatrix} - \begin{pmatrix}5/6\\1/3\\-1/6\end{pmatrix} \right\| \\
&= \left\| \begin{pmatrix}1/6\\-1/3\\-1/6\end{pmatrix} \right\| \\
&= \sqrt{(1/6)^2 + (-1/3)^2 + (-1/6)^2} \\
&= \frac{1}{\sqrt{6}}.
\end{aligned}$$



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Motivation

All of the procedures we learned require an *orthogonal* basis $\{u_1, u_2, \ldots, u_m\}$.

- Finding the $\mathcal{B}$-coordinates of a vector x using dot products:

$$x = \sum_{i=1}^{m} \frac{x \cdot u_i}{u_i \cdot u_i} u_i$$

- Finding the orthogonal projection of a vector x onto the span W of $u_1, u_2, \ldots, u_m$:

$$\text{proj}_W(x) = \sum_{i=1}^{m} \frac{x \cdot u_i}{u_i \cdot u_i} u_i.$$

Problem: What if your basis isn't orthogonal?

Solution: The Gram–Schmidt process: take any basis and make it orthogonal.

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The Gram–Schmidt Process

Let $\{v_1, v_2, \ldots, v_m\}$ be a basis for a subspace W of $\mathbf{R}^n$. Define:

1. $u_1 = v_1$
2. $u_2 = v_2 - \text{proj}_{\text{Span}\{u_1\}}(v_2) = v_2 - \dfrac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1$
3. $u_3 = v_3 - \text{proj}_{\text{Span}\{u_1, u_2\}}(v_3) = v_3 - \dfrac{v_3 \cdot u_1}{u_1 \cdot u_1} u_1 - \dfrac{v_3 \cdot u_2}{u_2 \cdot u_2} u_2$

$\vdots$

m. $u_m = v_m - \text{proj}_{\text{Span}\{u_1, u_2, \ldots, u_{m-1}\}}(v_m) = v_m - \displaystyle\sum_{i=1}^{m-1} \frac{v_m \cdot u_i}{u_i \cdot u_i} u_i$

Then $\{u_1, u_2, \ldots, u_m\}$ is an *orthogonal* basis for the same subspace W.

Remark

In fact, for every i between 1 and n, the set $\{u_1, u_2, \ldots, u_i\}$ is an orthogonal basis for $\text{Span}\{v_1, v_2, \ldots, v_i\}$.

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The Gram–Schmidt Process

Two vectors

Find an orthogonal basis $\{u_1, u_2\}$ for $W = \text{Span}\{v_1, v_2\}$, where

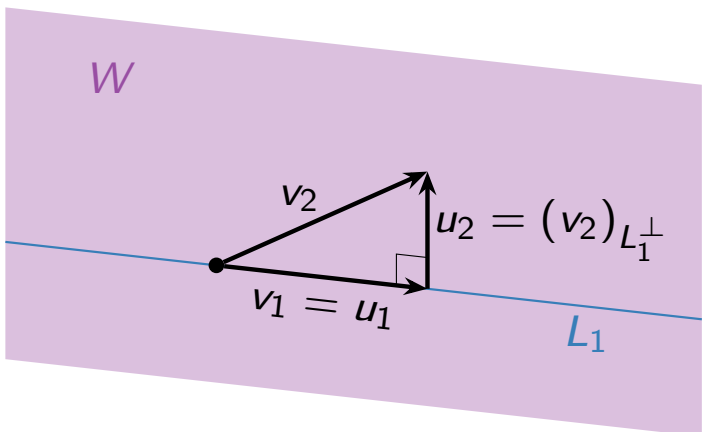
$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad v_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Run Gram–Schmidt:

$$1.\ u_1 = v_1 \qquad 2.\ u_2 = v_2 - \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{2}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Why does this work?

- First we take $u_1 = v_1$.
- Now we're sad because $u_1 \cdot v_2 \neq 0$, so we can't take $u_2 = v_2$.
- Fix: let $L_1 = \text{Span}\{u_1\}$, and let $u_2 = (v_2)_{L_1^\perp} = v_2 - \text{proj}_{L_1}(v_2)$.
- By construction, $u_1 \cdot u_2 = 0$, because $L_1 \perp u_2$.



Important: $\text{Span}\{u_1, u_2\} = \text{Span}\{v_1, v_2\} = W$: this is an *orthogonal* basis for the *same* subspace.

The Gram–Schmidt Process

Three vectors

Find an orthogonal basis $\{u_1, u_2, u_3\}$ for $W = \text{Span}\{v_1, v_2, v_3\} = \mathbf{R}^3$, where

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \qquad v_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \qquad v_3 = \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}.$$

Run Gram–Schmidt:

1. $u_1 = v_1$
2. $u_2 = v_2 - \dfrac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \dfrac{2}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$
3. $u_3 = v_3 - \dfrac{v_3 \cdot u_1}{u_1 \cdot u_1} u_1 - \dfrac{v_3 \cdot u_2}{u_2 \cdot u_2} u_2$
$$= \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} - \frac{4}{2} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{1} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

Important: $\text{Span}\{u_1, u_2, u_3\} = \text{Span}\{v_1, v_2, v_3\} = W$: this is an *orthogonal* basis for the *same* subspace.

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The Gram–Schmidt Process

Three vectors, continued

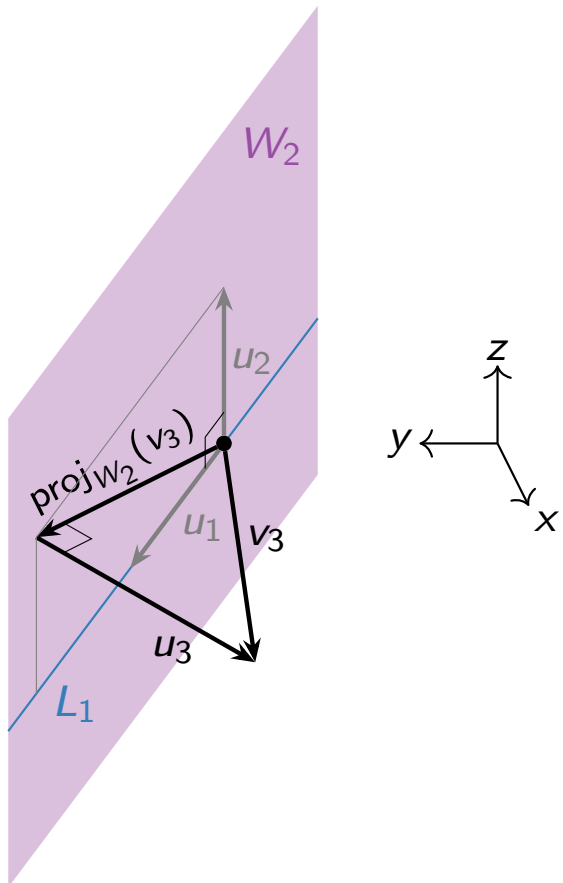
$$v_1 = \begin{pmatrix}1\\1\\0\end{pmatrix},\ v_2 = \begin{pmatrix}1\\1\\1\end{pmatrix},\ v_3 = \begin{pmatrix}3\\1\\1\end{pmatrix} \overset{\text{G–S}}{\rightsquigarrow} u_1 = \begin{pmatrix}1\\1\\0\end{pmatrix},\ u_2 = \begin{pmatrix}0\\0\\1\end{pmatrix},\ u_3 = \begin{pmatrix}1\\-1\\0\end{pmatrix}$$

Why does this work?

- Once we have u_1 and u_2, then we're sad because v_3 is not orthogonal to u_1 and u_2.
- Fix: let $W_2 = \text{Span}\{u_1, u_2\}$, and let $u_3 = (v_3)_{W_2^\perp} = v_3 - \text{proj}_{W_3}(u_3)$.
- By construction, $u_1 \cdot u_3 = 0 = u_2 \cdot u_3$ because $W_2 \perp u_3$.

Check:

$$u_1 \cdot u_2 = 0 \checkmark$$
$$u_1 \cdot u_3 = 0 \checkmark$$
$$u_2 \cdot u_3 = 0 \checkmark$$



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The Gram–Schmidt Process

Three vectors in $\mathbf{R}^4$

Find an orthogonal basis $\{u_1, u_2, u_3\}$ for $W = \text{Span}\{v_1, v_2, v_3\}$, where

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \qquad v_2 = \begin{pmatrix} -1 \\ 4 \\ 4 \\ -1 \end{pmatrix} \qquad v_3 = \begin{pmatrix} 4 \\ -2 \\ -2 \\ 0 \end{pmatrix}.$$

Run Gram–Schmidt:

1. $u_1 = v_1$
2. $u_2 = v_2 - \dfrac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1 = \begin{pmatrix} -1 \\ 4 \\ 4 \\ -1 \end{pmatrix} - \dfrac{6}{4} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -5/2 \\ 5/2 \\ 5/2 \\ -5/2 \end{pmatrix}$
3. $u_3 = v_3 - \dfrac{v_3 \cdot u_1}{u_1 \cdot u_1} u_1 - \dfrac{v_3 \cdot u_2}{u_2 \cdot u_2} u_2$

$$= \begin{pmatrix} 4 \\ -2 \\ -2 \\ 0 \end{pmatrix} - \frac{0}{24} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} - \frac{-20}{25} \begin{pmatrix} -5/2 \\ 5/2 \\ 5/2 \\ -5/2 \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \\ -2 \end{pmatrix}$$

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